Assignment X – Group Project: Back End

Physical Implementation of the Harris Corner Detection Accelerator (harris_top): Floorplan to GDSII

BEVD301P ASIC Design Lab · Fall Semester 2026–27 · Due 3 November 2026

Team Details

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1. Introduction

This report documents the back-end implementation of **harris_top**, our streaming Harris corner detection accelerator, from floorplanning to GDSII in Cadence Innovus. The input is the gate-level netlist and constraint file produced in the front end (Assignment IX): a 64 × 64-image configuration with 71,778 cells, synthesized in Genus at 2.25 ns and then constrained at 4 ns after the 2.25 ns target failed by 1.465 ns in the Sxy^2 multiplier. The whole back-end flow is scripted in innovus.tcl as one Tcl procedure per stage, with the multi-mode multi-corner (MMMC) timing setup in mmmc.tcl.

Result in brief: the routed design meets timing at 250 MHz (post-route setup WNS +0.307 ns, hold WNS +0.001 ns, no design-rule violations of capacitance, transition or fanout), passes Innovus DRC, connectivity and antenna checks with zero violations, and was streamed out to an 84 MB GDSII file.

Table 1: *Design and technology summary*

Item	Value
Top cell	harris_top_IMG_W64_IMG_H64 (harris_top with a 64 × 64 image)
Netlist	harris_top_netlist_slow.v (Genus, slow corner)
Constraints	harris_top_slow.sdc: 4.0 ns clock (250 MHz), input delay 0.06 ns, output delay 0.04 ns, max transition 0.5 ns, max fanout 32
Technology	Cadence 90 nm generic PDK, gsclib090 standard cells (gsclib090_translated.lef), setDesignMode -process 90, VDD = 0.9 V
Routing layers	Metal1 to Metal9
Power nets	VDD, VSS

Item	Value
Timing views	Setup: slow_view; hold: fast_view
Tool	Cadence Innovus Implementation System 21.15 (21.15-s110_1)

1.1 Scripted Flow

Each stage is a Tcl procedure that ends with saveDesign, and the next stage starts with restoreDesign of that checkpoint. A single stage can therefore be re-run after a change without repeating the earlier ones; the main section of the script calls the procedures in order, and stages that are already done can be commented out.

Table 2: *Stages of innovus.tcl*

Procedure	Main commands	Checkpoint
init_design_flow	init_design, checkDesign -all	01_init
floorplan	floorPlan, editPin	02_floorplan
powerplan	globalNetConnect, addRing, addStripe, sroute	03_power
placement	placeDesign, timeDesign -preCTS, optDesign -preCTS (+ -incr)	04_placement
cts	ccopt_design -cts, timeDesign / optDesign -postCTS (setup and hold)	05_cts
route	setNanoRouteMode, routeDesign -globalDetail	06_route
postroute	optDesign -postRoute (setup and hold), timeDesign, addFiller	07_postRoute
final	verify_drc, verifyConnectivity, verifyProcessAntenna, reports, saveNetlist, write_sdf, streamOut	08_final

2. Design Import and MMMC Setup

init_design reads the gsclib090 LEF, the Genus netlist and mmmc.tcl, with VDD and VSS as the global power nets. The MMMC file defines three library sets (slow, typical, fast), one constraint mode (func_mode, using harris_top_slow.sdc), a delay corner per library set and an analysis view per corner. checkDesign -all then checks the netlist against the libraries (missing cells, unconnected pins, floating nets) before any physical work starts.

Table 3: *MMMC analysis views (mmmc.tcl)*

Analysis view	Library	Used for	Reason
slow_view	slow.lib	Setup (active)	Slow process, low voltage and high temperature give the longest delays, the worst case for setup

Analysis view	Library	Used for	Reason
fast_view	fast.lib	Hold (active)	Fast corner gives the shortest delays, the worst case for hold
typical_view	typical.lib	Defined, not active	Available for nominal reporting

3. Floorplanning

```
floorPlan -site gsclib090site -r 1 0.6 10.15 10.15 10.15 10.15
```

Table 4: *Floorplan settings*

Parameter	Value	Reason
Row site	gsclib090site	Standard-cell row site of the library
Aspect ratio (H/W)	1.0 requested (0.995 after snapping to whole rows and sites)	Square core: pins are placed on all four sides, which keeps the average wire length short; there is no external shape constraint
Target core utilization	0.60 (60 %)	Leaves 40 % whitespace for buffers and resizing during optimisation, clock buffers, hold-fix cells and routing tracks around the wide multipliers and 64-bit buses
Core-to-IO margin	10.15 µm on all sides	Space for the VDD/VSS core ring: offset 1.8 + 2 × width 1.8 + spacing 1.8 = 7.2 µm, leaving about 3 µm clearance

Table 5: *Resulting floorplan dimensions (Specify Floorplan dialog)*

Dimension	Value
Core size (W × H)	996.73 × 991.80 µm
Die size (W × H)	1017.03 × 1012.10 µm
Core area	988,557 µm² (0.989 mm²)
Die area	1,029,336 µm² (1.029 mm²)
Initial core utilization	59.99 %

The die is the core plus the 10.15 µm margin on each side (996.73 + 2 × 10.15 = 1017.03 µm). For the 593,021 µm² of Genus cells, a 60 % utilization gives a core of about 0.99 mm².

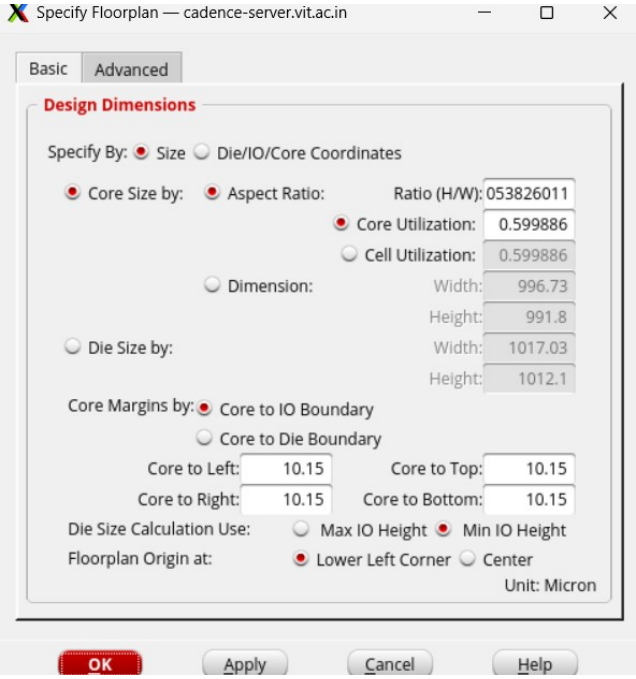


Figure 1: *Specify Floorplan dialog: core size, utilization and margins*

3.1 Pin Assignment

Pins were placed with editPin so that the layout follows the data flow: the pixel stream enters on the left and the results leave on the right, matching the left-to-right order of the pipeline. Pins on the top and bottom edges use Metal2 and pins on the left and right edges use Metal3, so each pin is on a layer whose preferred routing direction runs into the core from that edge.

Table 6: *Pin assignment*

Pins	Edge	Layer	Placement	Reason
clk	Top	Metal2	Centred	Clock enters at the middle of the edge, close to the centre of the core, which helps CTS keep insertion delay and skew low
k_q[10:0]	Top	Metal2	Spread along edge	Configuration input to harris_response
rst_n	Bottom	Metal2	Centred	Single control input
thr[63:0]	Bottom	Metal2	Spread along edge	Wide configuration bus to corner_threshold; a whole edge avoids crowding
pixel_valid, pixel_in[7:0]	Left	Metal3	Centred, 10 µm pitch	Input stream enters next to window3x3, the first stage
corner_out, corner_valid, resp_dbg[63:0]	Right	Metal3	Centred, 10 µm pitch	Outputs leave next to the last stages

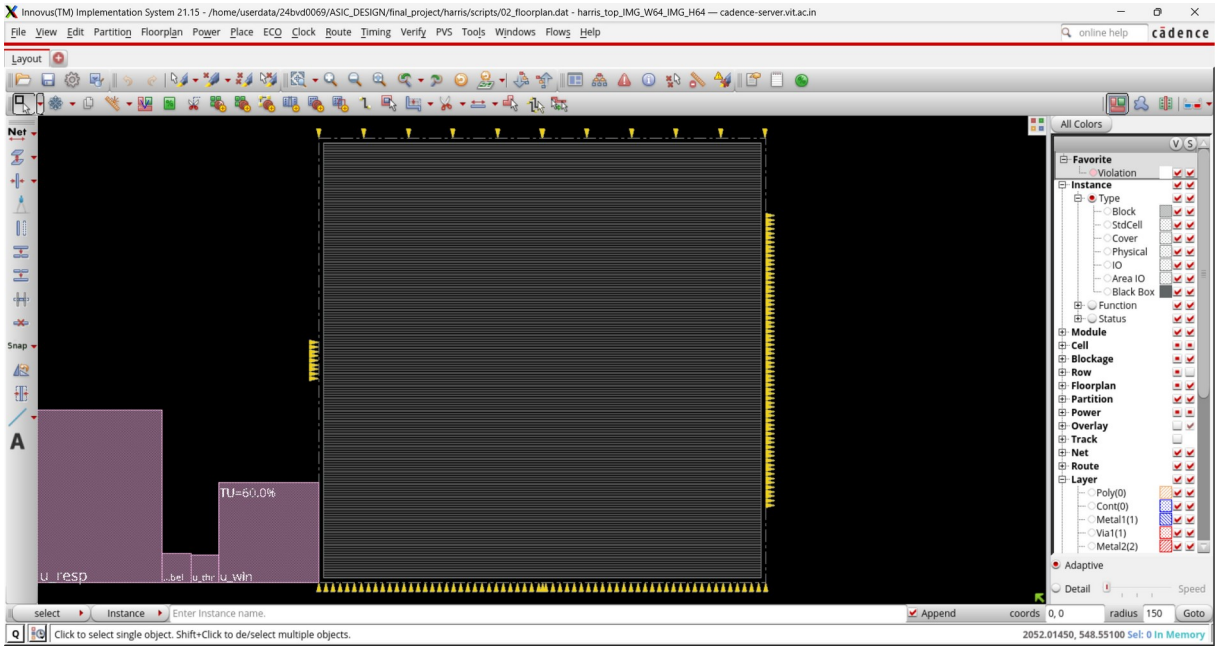


Figure 2: *Floorplan (02_floorplan): empty core rows and pins on all four sides; TU = 60 % is the target utilization*

4. Power Planning

```
globalNetConnect VDD -type pgpin -pin VDD -all
globalNetConnect VSS -type pgpin -pin VSS -all
addRing   -nets {VSS VDD} -type core_rings -follow core \
        -layer {top Metal3 bottom Metal3 left Metal4 right Metal4} -width 1.8 -spacing 1.8 -offset 1.8
addStripe -nets {VSS VDD} -layer Metal3 -direction horizontal -width 1.8 -spacing 1.8 -set_to_set_distance 60
addStripe -nets {VSS VDD} -layer Metal4 -direction vertical   -width 1.8 -spacing 1.8 -set_to_set_distance 60
sroute -connect {corePin} -nets {VDD VSS} -layerChangeRange {Metal1(1) Metal4(4)} ...
```

Table 7: *Power distribution network*

Element	Layer	Width	Spacing / pitch	Purpose
Global net connect	—	—	—	Ties the VDD/VSS pins of every cell to the VDD/VSS nets
Core ring (top, bottom)	Metal3	1.8 µm	1.8 µm spacing, 1.8 µm offset	Horizontal sides of the ring
Core ring (left, right)	Metal4	1.8 µm	1.8 µm spacing, 1.8 µm offset	Vertical sides of the ring
Horizontal stripes	Metal3	1.8 µm	VDD–VSS 1.8 µm, set-to-set 60 µm	Distribute power across the core
Vertical stripes	Metal4	1.8 µm	VDD–VSS 1.8 µm, set-to-set 60 µm	Together with Metal3 forms a mesh
Follow-pin rails (sroute)	Metal1	Cell rail	Every row	Connects the cell power pins along each row to the stripes and ring through vias up to Metal4

Reasons for these choices.

- Metal3 is used for horizontal and Metal4 for vertical straps, matching their preferred directions, so the ring and stripes form a regular mesh (visible as the grid in the power-plan view). Metal1 carries the cell rails and Metal2 stays free for cell pin access; Metal5 to Metal9 are left for signal routing.
- A 60 µm stripe pitch gives about 16 stripe pairs in each direction across the 997 µm core and keeps every cell within about 30 µm of a stripe, which limits IR drop along the Metal1 rails. Each VDD/VSS pair uses 1.8 + 1.8 + 1.8 = 5.4 µm per 60 µm, about 9 % of the Metal3 and Metal4 tracks, leaving the rest for signals.
- The ring and stripes use the same 1.8 µm width so the via arrays where they cross are uniform.
- `sroute` was added to the script to create the Metal1 rails of every row and connect them to the ring and stripes; without it the cells would have no power connection.

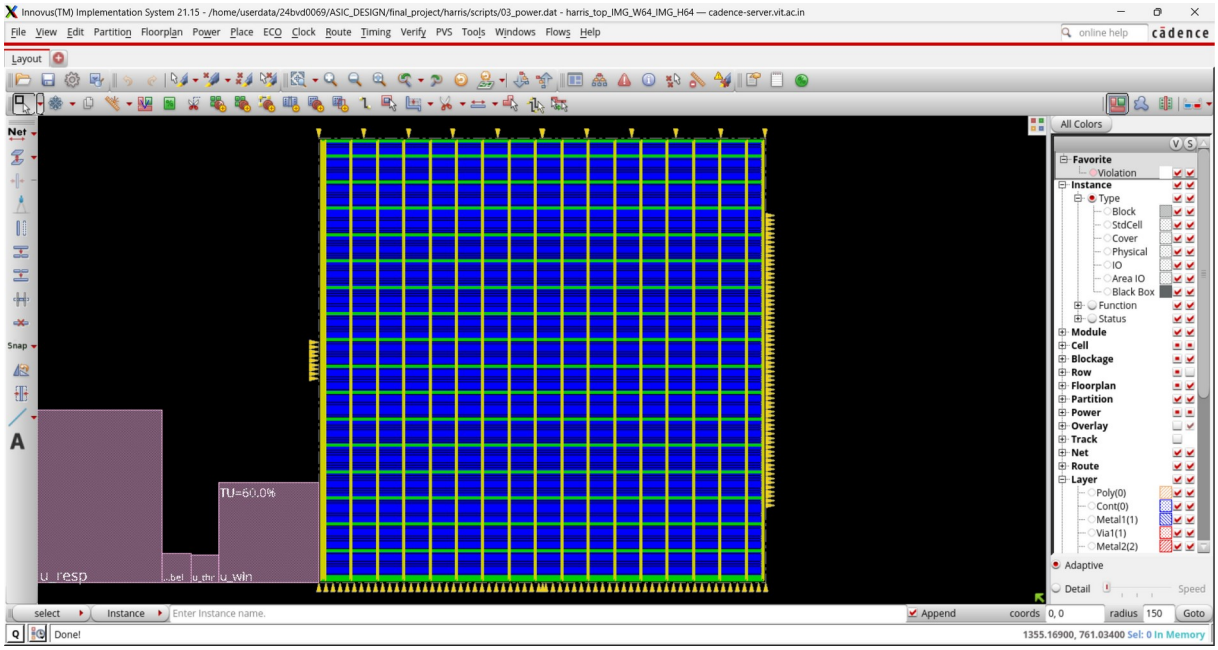


Figure 3: *Power plan (03_power): Metal3/Metal4 core ring and 60 µm stripe mesh with Metal1 rails*

5. Placement

```
placeDesign
timeDesign -preCTS -pathReports -drvReports -slackReports -numPaths 50 -outDir timingReports
setOptMode -fixCap true -fixTran true -fixFanoutLoad false
optDesign -preCTS
optDesign -preCTS -incr
```

- `placeDesign` places all standard cells in the rows.
- `timeDesign -preCTS` gives a baseline of setup slack and design-rule violations before optimisation. The clock is still ideal at this stage.
- `setOptMode` makes the optimiser fix max-capacitance and max-transition violations, which are real electrical problems. Fanout-load fixing is switched off:

max fanout is a guideline, and as long as transition and capacitance are within limits, buffering every high-fanout net (reset, enables, the thr bus) only adds area.

- optDesign -preCTS resizes cells, inserts buffers and restructures paths to fix setup and design-rule violations; the -incr run is a second, incremental pass on whatever remains.

Table 8: *Pre-CTS optimisation summary (ideal clock, 4 ns)*

Metric (after optDesign -preCTS -incr)	Value
Setup WNS / TNS (all paths)	−0.001 ns / −0.001 ns
Violating paths	1 (reg2reg) of 50,046
max_cap / max_tran / max_fanout / max_length violations	0 / 0 / 0 / 0
Density	56.918 %
Routing overflow (early global route)	0.00 % H, 0.00 % V

After placement and optimisation the design is almost closed even before CTS: a single register-to-register path misses by 1 ps, and there are no design-rule violations. The density fell from the 60 % target to 56.9 % because optimisation against the 4 ns constraint could downsize many cells that Genus had sized for 2.25 ns. The early global route shows zero overflow in both directions, so the 60 % utilization leaves enough routing resources and no congestion hot spots.

```
[24bvd0069@cadence-server timingReports]$ zcat harris_top_IMG_W64_IMG_H64_preCTS.summary.gz
############################################################
#  Generated by:      Cadence Innovus 21.15-s110_1
#  OS:                Linux x86_64(Host ID cadence-server.vit.ac.in)
#  Generated on:      Thu Oct 29 17:37:22 2026
#  Design:            harris_top_IMG_W64_IMG_H64
#  Command:           optDesign -preCTS -incr
############################################################

------------------------------------------------------------------
     optDesign Final Summary
------------------------------------------------------------------

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):| -0.001  | -0.001  |  0.004  |
|           TNS (ns):| -0.001  | -0.001  |  0.000  |
|    Violating Paths:|    1    |    1    |    0    |
|          All Paths:|  50046  |  49398  |  1783   |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 56.918%
Routing Overflow: 0.00% H and 0.00% V
------------------------------------------------------------------
[24bvd0069@cadence-server timingReports]$
```

Figure 4: *Pre-CTS optDesign summary (optDesign -preCTS -incr)*

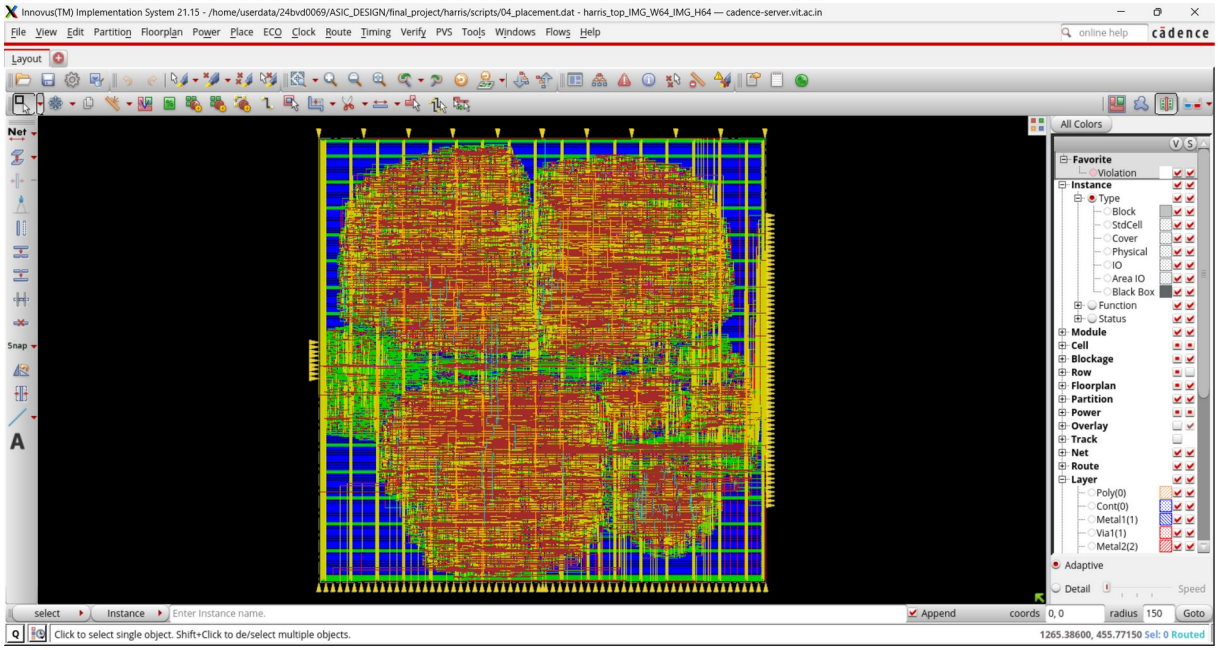


Figure 5: *Placed design (04_placement)*

The module view of the final layout shows how the placer grouped each block. u_win sits on the left next to the pixel input pins and u_resp at the lower right next to the output pins, while the three box-sum channels of u_box fill most of the core; u_sobel and u_prod form a narrow strip between them. This follows the left-to-right pin plan of Section 3.1 and keeps the long pipeline connections short.

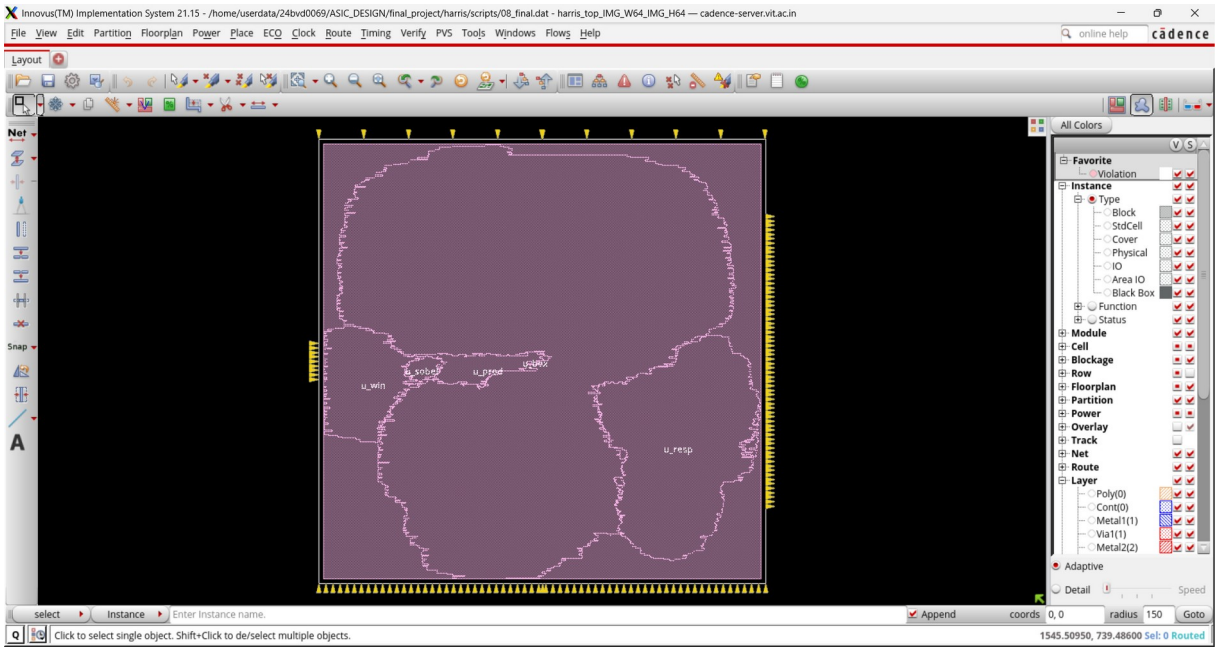


Figure 6: *Module (amoeba) view of the final layout: placement regions of u_win, u_sobel, u_prod, u_box and u_resp*

6. Clock Tree Synthesis (CTS)

```
set_ccopt_property target_max_trans 0.2
set_ccopt_property buffer_cells  { CLKBUFX2 CLKBUFX3 CLKBUFX4 CLKBUFX6 CLKBUFX8 CLKBUFX12
CLKBUFX16 CLKBUFX20 }
```

```
set_ccopt_property inverter_cells { CLKINVX1 CLKINVX2 CLKINVX3 CLKINVX4 CLKINVX6 CLKINVX8
CLKINVX12 CLKINVX16 CLKINVX20 }
ccopt_design -cts
report_ccopt_clock_trees -file cts_clock_trees.rpt
report_ccopt_skew_groups -file cts_skew_groups.rpt
```

- **Target clock transition 0.2 ns**, tighter than the 0.5 ns limit on data nets. Sharp clock edges reduce clock-to-Q delay variation and make skew easier to control.
- **Dedicated clock cells only.** CLKBUF and CLKINV cells have balanced rise and fall delays, which keeps the clock duty cycle and skew under control. The full range of drive strengths (X1 to X20) lets CCOpt size each level of the tree.
- `ccopt_design -cts` builds and balances the clock tree only; data-path timing is then fixed separately with `optDesign -postCTS` for setup and `optDesign -postCTS -hold` for hold.
- After CTS the clock is propagated, so setup and hold are analysed with real clock latencies. Hold only becomes meaningful at this point, which is why hold fixing starts after CTS.

Clock tree result. The clock net drives all 16,741 flip-flops, the largest fanout in the design. CCOpt built it from the CLKBUF cells (instances named CTS_ccl_a_buf_*, up to CLKBUFX16 and CLKBUFX20 in the final power report). The tree accounts for 13.47 mW, 10.6 % of the final power. Its effect on timing is visible in the final setup path (Section 8): the capture flip-flop receives the clock 1.062 ns after the launch flip-flop, which gives that path extra time. The layout after CTS looks the same as the placed design at full-chip zoom, since the clock buffers are small compared with the 54,000 standard cells, so no separate screenshot is included.

7. Routing

```
setNanoRouteMode -quiet -routeWithTimingDriven true
setNanoRouteMode -quiet -routeTdrEffort 1
setNanoRouteMode -quiet -routeTopRoutingLayer 9
setNanoRouteMode -quiet -routeBottomRoutingLayer 1
setNanoRouteMode -quiet -routeWithSiDriven false
routeDesign -globalDetail
```

Table 9: *NanoRoute settings*

Setting	Value	Reason
Timing-driven routing	true	Timing-critical nets are routed first and kept short
Timing-driven effort	1 (low)	Small timing weight, favouring routability
Routing layers	Metal1 to Metal9	All nine metals available, giving the 64-bit buses and multiplier nets enough tracks
SI-driven routing	false	Crosstalk-aware routing not used; crosstalk is not analysed in this flow
routeDesign -globalDetail	—	Global routing followed by detailed routing in one step

Change made during the project. The script originally set -drouteEndIteration 1, which stops detailed routing after its first iteration and leaves DRC violations unrepaired. The option was removed so NanoRoute runs its full search-and-repair loop; a duplicated routeWithTimingDriven line was also removed.

Routing result. Routing completed for all nets. The early global route had 0.00 % overflow in both directions, and after detailed routing the final checks report no DRC violations and no connectivity problems (no opens, shorts or unconnected pins; Section 9), so no congestion or routing violations remain.

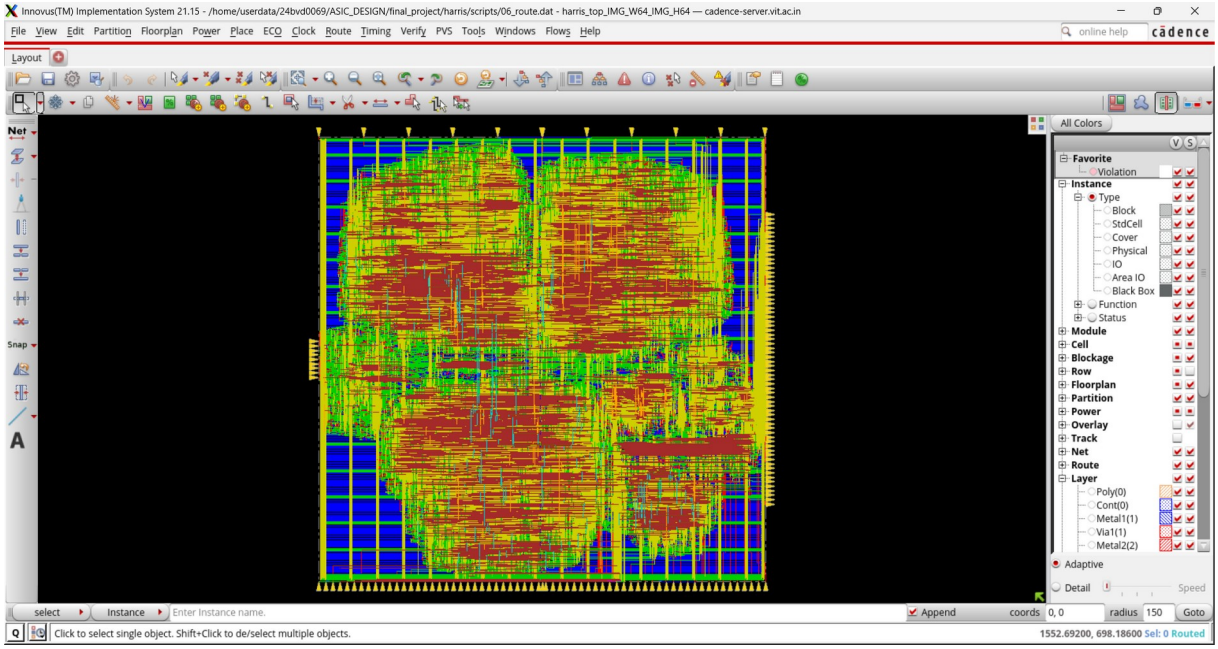


Figure 7: *Routed design (06_route)*

8. Timing Analysis and Closure

```
setAnalysisMode -analysisType onChipVariation -cppr both
optDesign -postRoute
optDesign -postRoute -hold
timeDesign -postRoute
timeDesign -postRoute -hold
addFiller -cell {FILL64 FILL32 FILL16 FILL8 FILL4 FILL2 FILL1} -prefix FILLER
```

- Post-route timing uses on-chip-variation analysis with common path pessimism removal (CPPR), which removes the false pessimism from the part of the clock tree shared by the launch and capture paths.
- Setup is optimised first and hold second, because hold fixing adds delay that must not reopen setup violations.

Table 10: *Timing across the flow (setup at slow corner, hold at fast corner)*

Stage	Clock	Setup WNS	Setup TNS	Setup viol.	Hold WNS	DRVs	Density
Genus (synthesis)	2.25 ns, ideal	−1.465 ns	−254.62 ns	378	—	—	—
Tempus (pre-layout)	3.8 ns, ideal	+0.049 ns	0	0	+0.054 ns	—	—
Pre-CTS (after opt)	4.0 ns, ideal	−0.001 ns	−0.001 ns	1	—	0	56.918 %
Post-route (final)	4.0 ns, propagated	+0.307 ns	0.000 ns	0	+0.001 ns	0	57.091 %

Timing improved at each step. The 1 ps pre-CTS violation was closed by the post-CTS and post-route optimisation, and the final design has +0.307 ns setup slack on all 50,046 paths with no violations of max capacitance, transition, fanout or length.

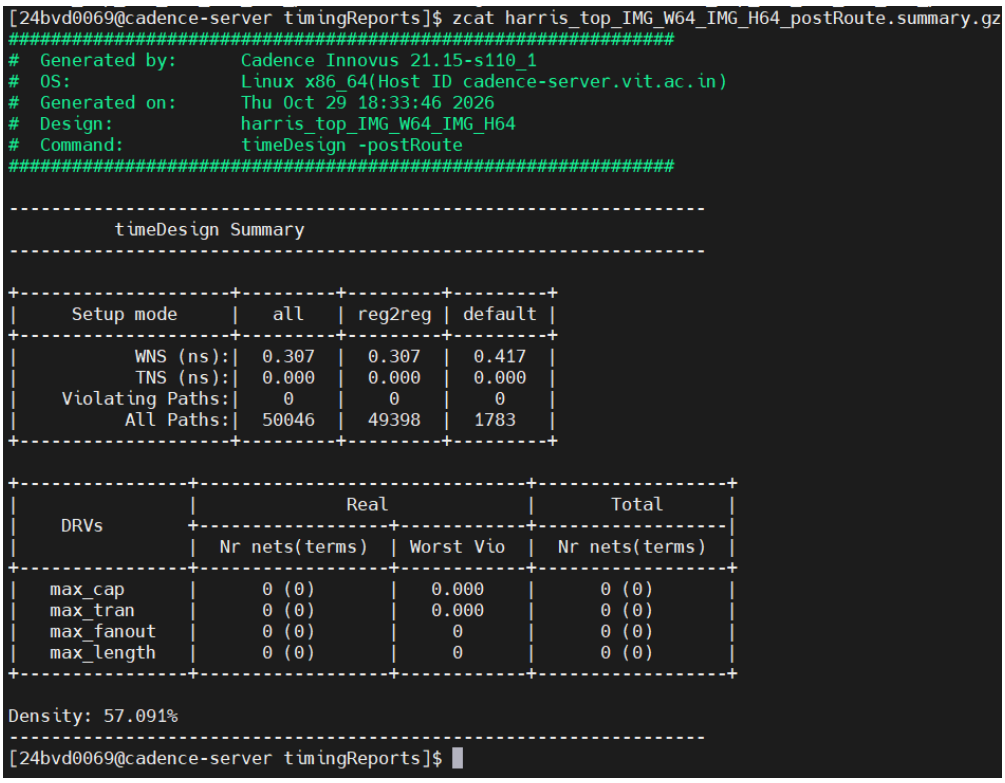


Figure 8: *Post-route timing summary (timeDesign -postRoute)*

Worst setup path

Table 11: *Final worst setup path (final_timing_setup.rpt)*

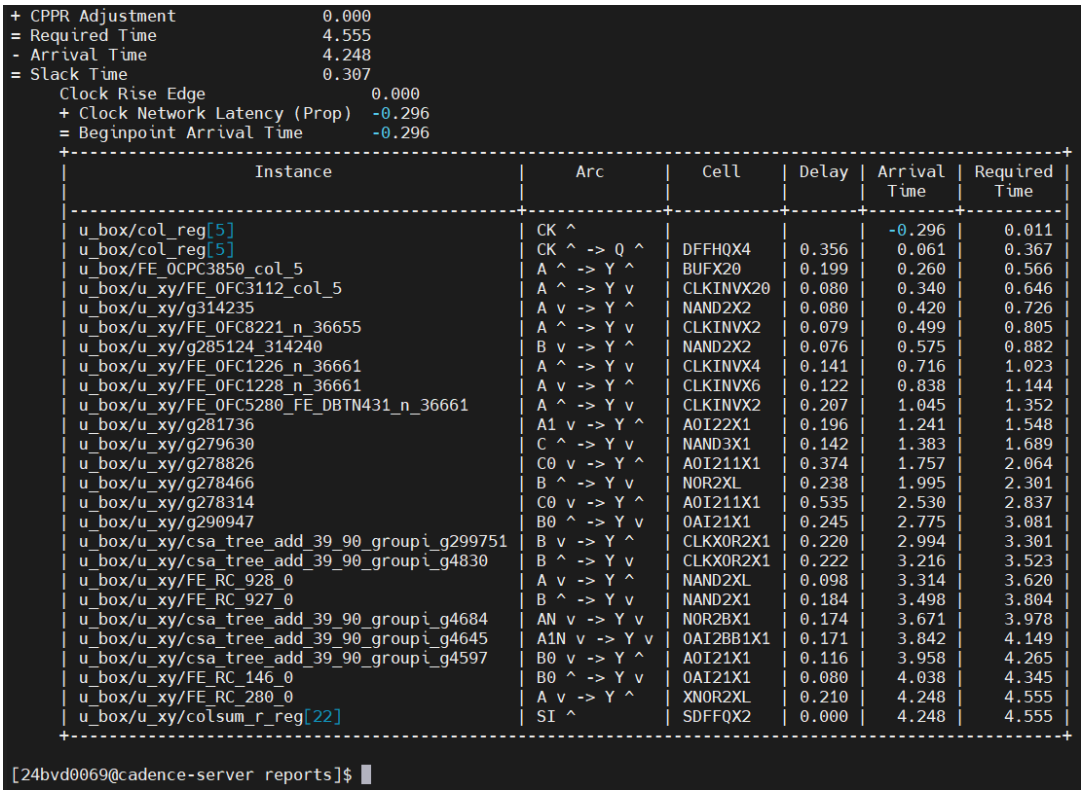
Item	Value
Analysis view	slow_view
Start point	u_box/col_reg[5] (column counter in boxsum5x5)
End point	u_box/u_xy/colsum_r_reg[22] (Sxy column-sum register)

Item	Value
Launch / capture clock latency	−0.296 ns / +0.766 ns
Setup time / required time	0.211 ns / 4.555 ns
Data arrival time	4.248 ns
Slack	+0.307 ns (MET)

After layout the worst path is no longer the Sxy² multiplier seen in synthesis but the column counter of boxsum5x5. The counter selects the line-buffer entry in every channel, so it is a high-fanout net: Innovus split it with buffers (BUFX20, CLKINVX20 and others), and the path then runs through the read logic and a carry-save adder tree into the Sxy column sum. The data path is about 4.54 ns long, longer than the 4 ns period, and meets timing only because the capture clock arrives 1.062 ns after the launch clock. This is the main path to improve if a faster clock is needed.

```
[24bvd0069@cadence-server reports]$ cat final_timing_setup.rpt
#############################################################
#  Generated by:      Cadence Innovus 21.15-s110_1
#  OS:                Linux x86_64(Host ID cadence-server.vit.ac.in)
#  Generated on:      Thu Oct 29 18:47:01 2026
#  Design:            harris_top_IMG_W64_IMG_H64
#  Command:           report_timing > ./reports/final_timing_setup.rpt
#############################################################
Path 1: MET Setup Check with Pin u_box/u_xy/colsum_r_reg[22]/CK
Endpoint:   u_box/u_xy/colsum_r_reg[22]/SI (^) checked with  leading edge of
'clk'
Beginpoint: u_box/col_reg[5]/Q             (^) triggered by  leading edge of
'clk'
Path Groups: {clk}
Analysis View: slow_view
Other End Arrival Time          0.766
- Setup                         0.211
+ Phase Shift                   4.000
+ CPPR Adjustment               0.000
= Required Time                 4.555
- Arrival Time                  4.248
= Slack Time                    0.307
     Clock Rise Edge                     0.000
     + Clock Network Latency (Prop)     -0.296
     = Beginpoint Arrival Time          -0.296
     +-------------------------------------------------------------------------------------------------+
     |                  Instance                  |     Arc      |   Cell    | Delay | Arrival | Required |
     |                                            |              |           |       |  Time   |   Time   |
     |--------------------------------------------+--------------+-----------+-------+---------+----------|
     | u_box/col_reg[5]                           | CK ^         |           |       |  -0.296 |    0.011 |
     | u_box/col_reg[5]                           | CK ^ -> Q ^  | DFFHQX4   | 0.356 |   0.061 |    0.367 |
     | u_box/FE_OCPC3850_col_5                    | A ^ -> Y ^   | BUFX20    | 0.199 |   0.260 |    0.566 |
     | u_box/u_xy/FE_OFC3112_col_5                | A ^ -> Y v   | CLKINVX20 | 0.080 |   0.340 |    0.646 |
     | u_box/u_xy/g314235                         | A v -> Y ^   | NAND2X2   | 0.080 |   0.420 |    0.726 |
     | u_box/u_xy/FE_OFC8221_n_36655              | A ^ -> Y v   | CLKINVX2  | 0.079 |   0.499 |    0.805 |
     | u_box/u_xy/g285124_314240                  | B v -> Y ^   | NAND2X2   | 0.076 |   0.575 |    0.882 |
     | u_box/u_xy/FE_OFC1226_n_36661              | A ^ -> Y v   | CLKINVX4  | 0.141 |   0.716 |    1.023 |
     | u_box/u_xy/FE_OFC1228_n_36661              | A v -> Y ^   | CLKINVX6  | 0.122 |   0.838 |    1.144 |
     | u_box/u_xy/FE_OFC5280_FE_DBTN431_n_36661   | A ^ -> Y v   | CLKINVX2  | 0.207 |   1.045 |    1.352 |
     | u_box/u_xy/g281736                         | A1 v -> Y ^  | AOI22X1   | 0.196 |   1.241 |    1.548 |
```

Figure 9: *Final setup report: worst path summary*



```
+ CPPR Adjustment              0.000
= Required Time                4.555
- Arrival Time                 4.248
= Slack Time                   0.307
     Clock Rise Edge                0.000
     + Clock Network Latency (Prop)  -0.296
     = Beginpoint Arrival Time       -0.296
     +-------------------------------------------------------------------------------------------------------+
     |                  Instance                  |     Arc      |   Cell    | Delay | Arrival | Required |
     |                                            |              |           |       |  Time   |   Time   |
     |--------------------------------------------+--------------+-----------+-------+---------+----------|
     | u_box/col_reg[5]                           | CK ^         |           |       |  -0.296 |    0.011 |
     | u_box/col_reg[5]                           | CK ^ -> Q ^  | DFFHQX4   | 0.356 |   0.061 |    0.367 |
     | u_box/FE_OCPC3850_col_5                    | A ^ -> Y ^   | BUFX20    | 0.199 |   0.260 |    0.566 |
     | u_box/u_xy/FE_OFC3112_col_5                | A ^ -> Y v   | CLKINVX20 | 0.080 |   0.340 |    0.646 |
     | u_box/u_xy/g314235                         | A v -> Y ^   | NAND2X2   | 0.080 |   0.420 |    0.726 |
     | u_box/u_xy/FE_OFC8221_n_36655              | A ^ -> Y v   | CLKINVX2  | 0.079 |   0.499 |    0.805 |
     | u_box/u_xy/g285124_314240                  | B v -> Y ^   | NAND2X2   | 0.076 |   0.575 |    0.882 |
     | u_box/u_xy/FE_OFC1226_n_36661              | A ^ -> Y v   | CLKINVX4  | 0.141 |   0.716 |    1.023 |
     | u_box/u_xy/FE_OFC1228_n_36661              | A v -> Y ^   | CLKINVX6  | 0.122 |   0.838 |    1.144 |
     | u_box/u_xy/FE_OFC5280_FE_DBTN431_n_36661   | A ^ -> Y v   | CLKINVX2  | 0.207 |   1.045 |    1.352 |
     | u_box/u_xy/g281736                         | A1 v -> Y ^  | AOI22X1   | 0.196 |   1.241 |    1.548 |
     | u_box/u_xy/g279630                         | C ^ -> Y v   | NAND3X1   | 0.142 |   1.383 |    1.689 |
     | u_box/u_xy/g278826                         | C0 v -> Y ^  | AOI211X1  | 0.374 |   1.757 |    2.064 |
     | u_box/u_xy/g278466                         | B ^ -> Y v   | NOR2XL    | 0.238 |   1.995 |    2.301 |
     | u_box/u_xy/g278314                         | C0 v -> Y ^  | AOI211X1  | 0.535 |   2.530 |    2.837 |
     | u_box/u_xy/g290947                         | B0 ^ -> Y v  | OAI21X1   | 0.245 |   2.775 |    3.081 |
     | u_box/u_xy/csa_tree_add_39_90_groupi_g299751 | B v -> Y ^ | CLKXOR2X1 | 0.220 |   2.994 |    3.301 |
     | u_box/u_xy/csa_tree_add_39_90_groupi_g4830 | B ^ -> Y v   | CLKXOR2X1 | 0.222 |   3.216 |    3.523 |
     | u_box/u_xy/FE_RC_928_0                     | A v -> Y ^   | NAND2XL   | 0.098 |   3.314 |    3.620 |
     | u_box/u_xy/FE_RC_927_0                     | B ^ -> Y v   | NAND2X1   | 0.184 |   3.498 |    3.804 |
     | u_box/u_xy/csa_tree_add_39_90_groupi_g4684 | AN v -> Y v  | NOR2BX1   | 0.174 |   3.671 |    3.978 |
     | u_box/u_xy/csa_tree_add_39_90_groupi_g4645 | A1N v -> Y v | OAI2BB1X1 | 0.171 |   3.842 |    4.149 |
     | u_box/u_xy/csa_tree_add_39_90_groupi_g4597 | B0 v -> Y ^  | AOI21X1   | 0.116 |   3.958 |    4.265 |
     | u_box/u_xy/FE_RC_146_0                     | B0 ^ -> Y v  | OAI21X1   | 0.080 |   4.038 |    4.345 |
     | u_box/u_xy/FE_RC_280_0                     | A v -> Y ^   | XNOR2XL   | 0.210 |   4.248 |    4.555 |
     | u_box/u_xy/colsum_r_reg[22]                | SI ^         | SDFFQX2   | 0.000 |   4.248 |    4.555 |
     +-------------------------------------------------------------------------------------------------------+

[24bvd0069@cadence-server reports]$
```

Figure 10: *Final setup report: full worst path*

Worst hold path

The worst hold path (fast_view) runs from the input pin pixel_in[3] into the window3x3 line buffer (u_win/lb0_reg[40][3]). With only 0.06 ns of input delay and the capture clock arriving 0.244 ns after the edge, the data would arrive too early, so post-route hold optimisation inserted delay cells on the input (FE_PHC8932, a DLY3X1, and FE_PHC8928, a CLKBUFX6). The path now meets hold with +0.001 ns slack. All hold paths are met.

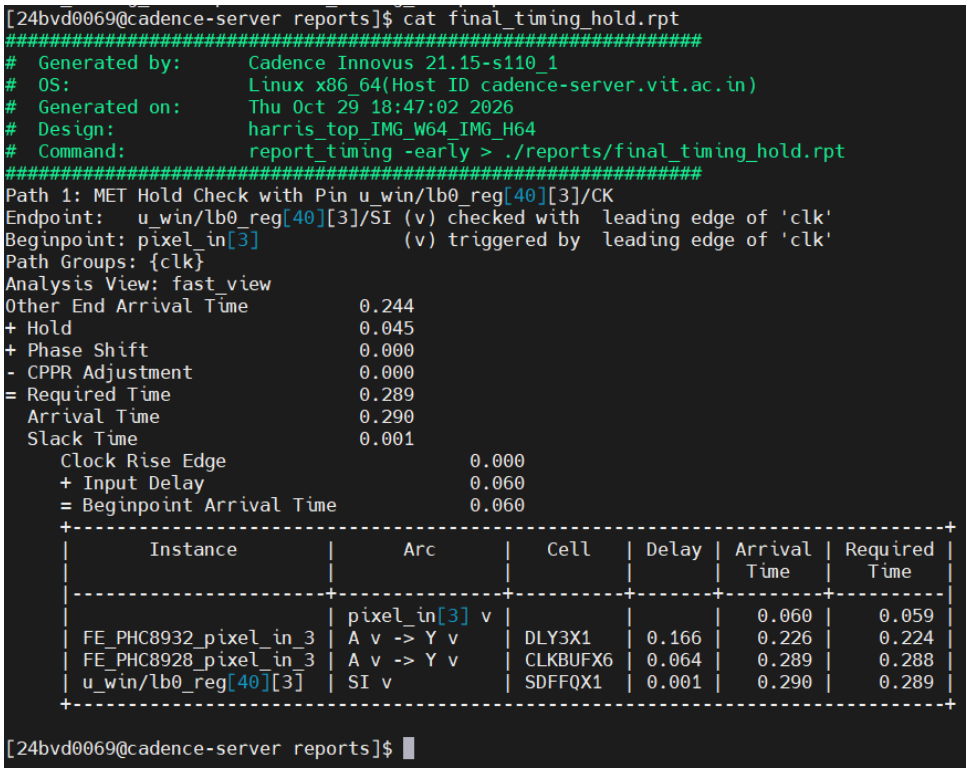


Figure 11: *Final hold report: worst path with the inserted hold-fix delay cells*

9. Physical Verification and GDSII

```
verify_drc -report ./reports/final_drc.rpt
verifyConnectivity -type all -report ./reports/final_connectivity.rpt
verifyProcessAntenna -report ./reports/final_antenna.rpt
saveNetlist ./output/harris_top_postroute.v
write_sdf   ./output/harris_top_postroute.sdf
streamOut ./output/harris_top.gds -libName harris_top \
   -merge .../gds/gsclib090.gds -units 1000 -mode ALL
```

Table 12: *Physical verification results*

Check	Command	What it checks	Result
DRC	verify_drc	Spacing, width, enclosure and short violations on all layers	No DRC violations found
Connectivity	verifyConnectivity -type all	Opens, unconnected pins, dangling wires and loops against the netlist	No problems or warnings
Process antenna	verifyProcessAntenna	Charge build-up on long wires connected to gate inputs during fabrication	No violations found
LVS	—	Layout against schematic with a sign-off tool	Not part of our flow (see below)

There are **no unresolved DRC, connectivity or antenna violations**. A separate LVS run with a sign-off tool (such as Assura, PVS or Calibre) was not part of the flow available to us. The connectivity check is the closest equivalent inside Innovus: it confirms that every net in the layout is connected exactly as in the netlist. The post-

layout netlist (harris_top_postroute.v) and SDF were written out for LVS and post-layout simulation.

```
[24bvd0069@cadence-server reports]$ cat final_drc.rpt
############################################################
#  Generated by:      Cadence Innovus 21.15-s110_1
#  OS:                Linux x86_64(Host ID cadence-server.vit.ac.in)
#  Generated on:      Thu Oct 29 18:46:10 2026
#  Design:            harris_top_IMG_W64_IMG_H64
#  Command:           verify_drc -report ./reports/final_drc.rpt
############################################################

No DRC violations were found

[24bvd0069@cadence-server reports]$
```

```
[24bvd0069@cadence-server reports]$ cat final_connectivity.rpt
############################################################
#  Generated by:      Cadence Innovus 21.15-s110_1
#  OS:                Linux x86_64(Host ID cadence-server.vit.ac.in)
#  Generated on:      Thu Oct 29 18:46:25 2026
#  Design:            harris_top_IMG_W64_IMG_H64
#  Command:           verifyConnectivity -type all -report ./reports/final_connectivity.rpt
############################################################
Verify Connectivity Report is created on Thu Oct 29 18:46:25 2026



Begin Summary
    Found no problems or warnings.
End Summary
[24bvd0069@cadence-server reports]$
```

```
[24bvd0069@cadence-server reports]$ cat final_antenna.rpt
############################################################
#  Generated by:      Cadence Innovus 21.15-s110_1
#  OS:                Linux x86_64(Host ID cadence-server.vit.ac.in)
#  Generated on:      Thu Oct 29 18:46:27 2026
#  Design:            harris_top_IMG_W64_IMG_H64
#  Command:           verifyProcessAntenna -report ./reports/final_antenna.rpt
############################################################

No Violations Found
[24bvd0069@cadence-server reports]$
```

Figure 12: *Final DRC, connectivity and antenna reports*

GDSII stream-out. streamOut wrote the final layout to **harris_top.gds (84 MB, 29 Oct 2026)** in the output folder, next to the post-layout netlist and SDF. The gsclib090 cell GDS is merged in, so the standard cells appear with their full layout rather than as abstracts; -units 1000 sets 1000 database units per micron (1 nm resolution) and -mode ALL writes all layout data.

```
[24bvd0069@cadence-server scripts]$ cd output/
[24bvd0069@cadence-server output]$ ls
harris_top.gds  harris_top_postroute.sdf  harris_top_postroute.v
[24bvd0069@cadence-server output]$ ls -lh h
harris_top.gds            harris_top_postroute.sdf  harris_top_postroute.v
[24bvd0069@cadence-server output]$ ls -lh harris_top.gds
-rw-rw-r--. 1 24bvd0069 24bvd0069 84M Oct 29 18:47 harris_top.gds
[24bvd0069@cadence-server output]$
```

Figure 13: *Output folder with the GDSII file (84 MB), post-layout netlist and SDF*

10. Discussion of Results

10.1 Final Implementation Metrics

Table 13: *Final metrics*

Metric	Value
Clock period / frequency	4.0 ns / 250 MHz
Throughput	250 Mpixel/s (1 pixel per clock)
Die size / area	1017.03 × 1012.10 µm / 1.029 mm²
Core size / area	996.73 × 991.80 µm / 0.989 mm²
Standard-cell area	564,377 µm²
Instances	54,154
Final density	57.09 %
Setup WNS / hold WNS	+0.307 ns / +0.001 ns
Total power	127.30 mW (internal 97.06, switching 26.79, leakage 3.45)
DRC / connectivity / antenna	0 / 0 / 0
GDSII	harris_top.gds, 84 MB

Area

Table 14: *Final area by instance (final_area.rpt) compared with synthesis*

Instance	Module	Instances	Area (µm²)	Share	vs. Genus
(top)	harris_top_IMG_W64_IMG_H64	54,154	564,377.19	100 %	−4.8 %
u_box	boxsum5x5	36,099	435,602.76	77.18 %	−4.3 %
u_xy	boxsum_ch (21-bit)	12,781	154,763.34	27.42 %	
u_xx	boxsum_ch (20-bit)	11,651	139,930.37	24.79 %	
u_yy	boxsum_ch (20-bit)	11,540	139,701.03	24.75 %	
u_resp	harris_response	14,528	86,555.30	15.34 %	−7.3 %
u_win	window3x3	1,820	29,579.65	5.24 %	−5.5 %
u_prod	gradient_products	1,082	7,024.28	1.24 %	−14.0 %
u_thr	corner_threshold	310	2,401.64	0.43 %	+3.1 %
u_sobel	sobel_grad	262	2,395.59	0.42 %	−11.2 %

The final cell area is 4.8 % smaller than after synthesis and the instance count is 24.6 % lower (54,154 vs. 71,778), even though Innovus added clock buffers and hold-fix cells. This is consistent with the netlist having been synthesized for 2.25 ns: against the relaxed 4 ns constraint, Innovus optimisation could downsize cells and remove

buffering that was no longer needed. The distribution between modules is unchanged, with the box-sum line buffers taking 77 % of the area.

```
[24bvd0069@cadence-server reports]$ cat final_area.rpt
    Hinst Name            Module Name                              Inst Count          Total Area
-------------------------------------------------------------------------------------------------
harris_top_IMG_W64_IMG_H64                                             54154          564377.187
 u_box                    boxsum5x5_IMG_W62_IMG_H62                    36099          435602.762
  u_box/u_xx              boxsum_ch_IN_W20_SGN0_IMG_W62_CWID6          11651          139930.374
  u_box/u_xy              boxsum_ch_IN_W21_SGN1_IMG_W62_CWID6          12781          154763.343
  u_box/u_yy              boxsum_ch_IN_W20_SGN0_IMG_W62_CWID6_673      11540          139701.033
 u_prod                   gradient_products                             1082            7024.789
 u_resp                   harris_response_K_FRAC10_R_W64               14528           86555.300
 u_sobel                  sobel_grad                                     262            2395.588
 u_thr                    corner_threshold_R_W64                         310            2401.644
 u_win                    window3x3_IMG_W64_IMG_H64                     1820           29579.652
[24bvd0069@cadence-server reports]$
```

Figure 14: *Final area report by instance*

Power

Table 15: *Final power (report_power, VDD = 0.9 V, 4 ns clock)*

Group	Internal (mW)	Switching (mW)	Leakage (mW)	Total (mW)	Share
Sequential	74.06	2.963	2.285	79.31	62.3 %
Combinational	18.88	14.51	1.127	34.52	27.12 %
Clock (combinational)	4.112	9.317	0.041	13.47	10.58 %
Total	**97.06**	**26.79**	**3.453**	**127.30**	**100 %**

Final power is 127.3 mW, compared with Genus's 198.7 mW estimate at 2.25 ns; the reduction mainly reflects the lower clock frequency (250 MHz instead of 444 MHz) and the smaller cell area. As in synthesis, the flip-flops dominate (62 %), and the clock tree adds 10.6 %. The figure uses default switching activity rather than simulation data, so it is an estimate.

```
Total Power
--------------------------------------------------------------------------------------
Total Internal Power:       97.05618587           76.2410%
Total Switching Power:      26.79280971           21.0467%
Total Leakage Power:         3.45281997            2.7123%
Total Power:               127.30181554
--------------------------------------------------------------------------------------


Group                           Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
--------------------------------------------------------------------------------------
Sequential                         74.06       2.963       2.285       79.31        62.3
Macro                                 0           0           0           0           0
IO                                    0           0           0           0           0
Combinational                      18.88       14.51       1.127       34.52       27.12
Clock (Combinational)              4.112       9.317      0.0409       13.47       10.58
Clock (Sequential)                    0           0           0           0           0
--------------------------------------------------------------------------------------
Total                              97.06       26.79       3.453       127.3         100
--------------------------------------------------------------------------------------


Rail                 Voltage   Internal   Switching     Leakage       Total  Percentage
                               Power      Power         Power         Power  (%)
--------------------------------------------------------------------------------------
VDD                      0.9      97.06       26.79       3.453       127.3         100


Clock                           Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
--------------------------------------------------------------------------------------
clk                                4.112       9.317      0.0409       13.47       10.58
--------------------------------------------------------------------------------------
Total                              4.112       9.317      0.0409       13.47       10.58
--------------------------------------------------------------------------------------
Clock: clk
Clock Period: 0.004000 usec
Clock Toggle Rate:   500.0000 Mhz
Clock Static Probability:  0.2813
```

Figure 15: *Final power report*

10.2 Does the design meet its goals?

- **Timing.** Yes. At the 4 ns (250 MHz) constraint the post-route setup WNS is +0.307 ns and hold WNS is +0.001 ns, with no violating paths and no design-rule (cap, transition, fanout, length) violations.
- **DRC and LVS.** verify_drc, verifyConnectivity and verifyProcessAntenna all report zero violations. No sign-off LVS was run (Section 9).
- **Area.** 564,377 µm² of cells in a 0.989 mm² core (1.029 mm² die) at 57.1 % density.
- **Output.** The final GDSII (harris_top.gds, 84 MB), post-layout netlist and SDF were generated.

10.3 Issues Encountered

Table 16: *Back-end issues and fixes*

Issue	Resolution
Setup failed by 1.465 ns at the 2.25 ns synthesis clock (Sxy² multiplier, see Assignment IX)	Clock relaxed to 4 ns in the SDC used for implementation
One path missed setup by 1 ps after pre-CTS optimisation	Closed by post-CTS and post-route optimisation; final WNS +0.307 ns
Input-to-line-buffer paths too fast for hold at the fast corner	Post-route hold fixing inserted delay cells (DLY3X1); final hold WNS +0.001 ns
Detailed routing left DRC violations	Removed -drouteEndIteration 1 so NanoRoute completes its repair iterations; final DRC clean

Issue	Resolution
Cell rows not connected to the power grid	Added sroute to create the Metal1 rails and connect them to the ring and stripes
Parameterised top cell renamed by Genus	restoreDesign and init_design use harris_top_IMG_W64_IMG_H64

10.4 Remaining Limitations

- **Filler cells.** The script calls addFiller, but the final power report counts 0 fillers in the saved design, so the gaps between cells are still empty. Innovus DRC passes without them, but they would be needed for well and rail continuity in a sign-off DRC/LVS run.
- **Clock waveform.** When the period was changed to 4 ns, the falling edge in the SDC was left at 1.125 ns from the original 2.25 ns clock, giving a 28 % duty cycle (the power report's clock static probability of 0.2813 reflects this). Timing is unaffected because every flip-flop uses the rising edge, but the waveform should be {0 2}.
- **Timing margin.** Hold slack is only +0.001 ns, and the worst setup path depends on 1.06 ns of favourable clock skew.
- **Sign-off accuracy.** No RC corner (QRC or capacitance table) is defined in mmmc.tcl, so parasitics come from the LEF layer data; setup is checked only at the slow corner and hold only at the fast corner; crosstalk, IR drop and electromigration were not analysed; power uses default switching activity; and no sign-off LVS was run.
- **Architecture.** The line buffers are flip-flops (77 % of the area, 62 % of the power). SRAM macros or clock gating would reduce both for larger images.